Slide 1: Introduction to Computers and Networks



Slide 2: OUTLINE

- Computer Evolution and generations
- Hardware & Software
- Operating Systems
- Computer Components
- Memory Hierarchy
- GPU
- Computer Networks
- Network Hardware
- Protocols
- The Internet
- Browsers

Slide 3: COMPUTER EVOLUTION

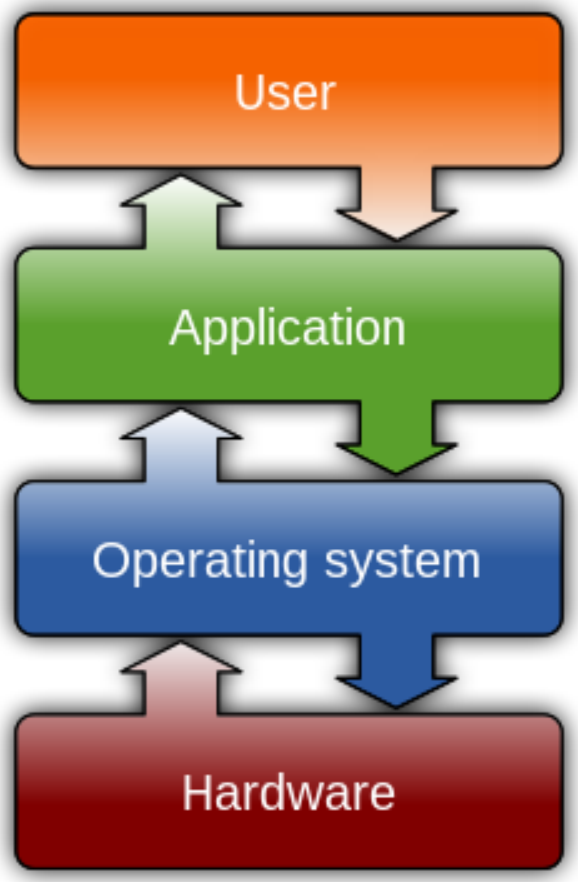
- First Generation Vacuum Tubes (1940 – 1956)--vacuum tube
- Second Generation Transistors (1956 – 1963)--transistor
- Third Generation Integrated Circuits (1964 – 1971)--integrated circuit (IC)
- Fourth Generation Microprocessors (1971 – present)--microprocessor
- Fifth Generation Artificial Intelligence (Present & Beyond)--integration of AI technologies into computers.

Slide 4: HARDWARE & SOFTWARE

- Hardware refers to the physical component of a computer.
- Hardware are devices required to store and execute the software.
- Hardware serves as a delivery system for software solutions.
- Software is a collection of instructions and procedures that perform different tasks on the computer to deliver a set of instructions, software is installed on hardware.

Slide 5: OPERATING SYSTEM

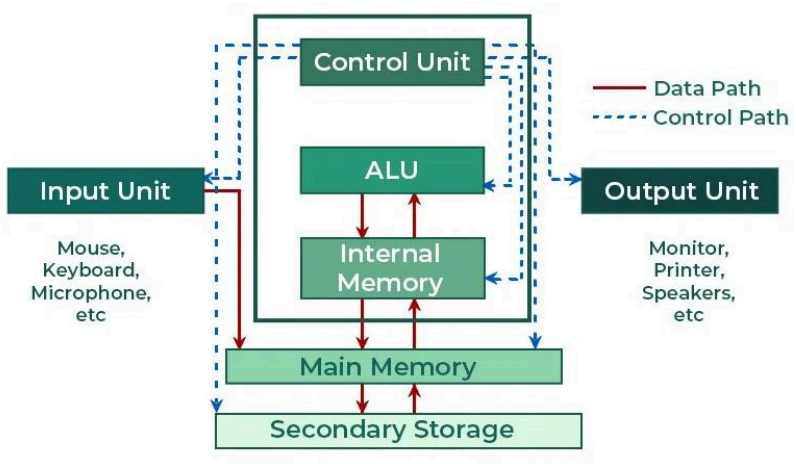
- Manages computer hardware and software resources.
- Starts when the computer boots.
- Provides a user-friendly interface and translates user commands for the hardware.
- Controls devices such as the keyboard, mouse, monitor, and storage.
- Examples: Windows, Ubuntu Linux, and macOS.



Slide 6: COMPUTER COMPONENTS

A computer has five basic components:

- Input unit - enters data and instructions.
- Output unit - presents results.
- Memory unit - stores data and instructions.
- Control unit - directs computer operations.
- Arithmetic and logic unit (ALU) - performs calculations and comparisons.



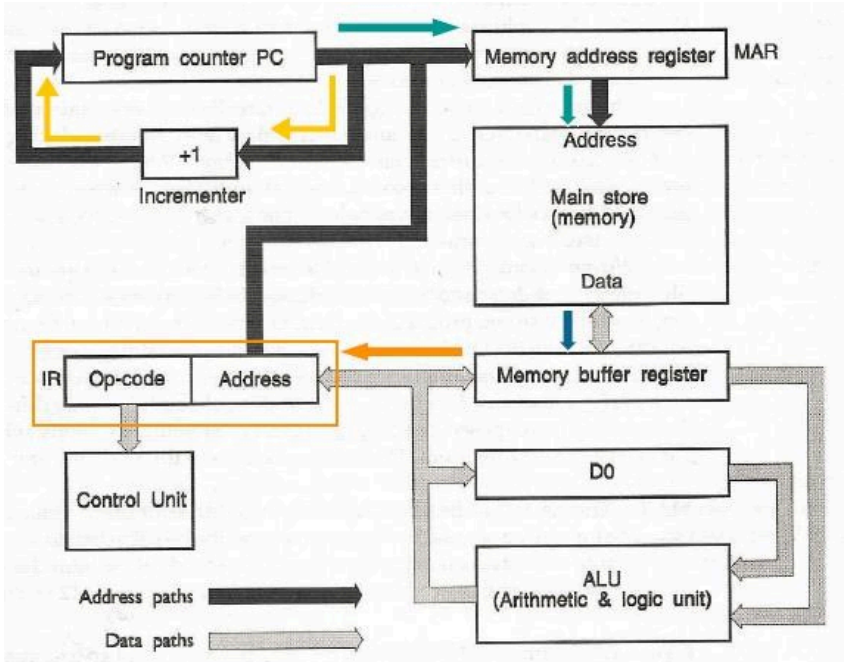
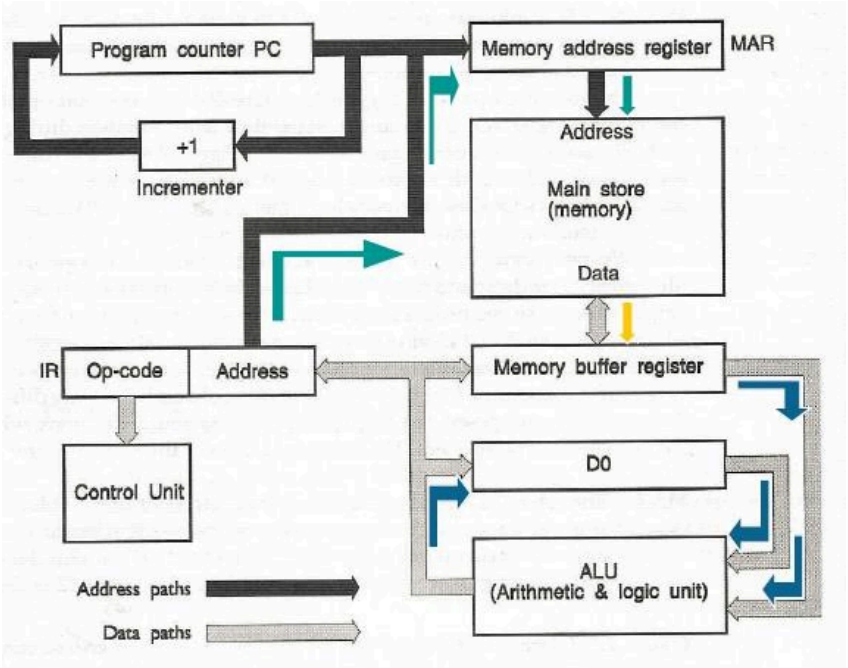
Slide 7: CPU EVOLUTION

- Intel CPU generations:
  - 8086
  - 80x86
  - Pentium
  - Pentium MMX
  - Pentium II, Pentium III, Celeron
  - Pentium IV, Centrino
  - Dual Core, Core 2 Duo, Core 2 Quad
  - Core i3, i5, i7, i9
  - Core Ultra 9

Slide 8: MACHINE CYCLE

- The machine cycle consists of 4 main steps repeated for every instruction your computer carries out.
    - Fetch
    - Decode
    - Execute
    - Store

Slide 9: FETCH AND EXECUTE

Fetch		Execute	

Slide 10: MAIN MEMORY

- The main memory only holds data and instructions on which the computer is currently working.
- It is divided into two subcategories:
  - Random Access Memory (RAM)
  - Read Only Memory (ROM)

Slide 11: Random Access Memory (RAM) and Read Only Memory (ROM)

- Volatile memory
- Used to store the operating system, application program, and data that the computer is currently using
- Exists as a chip on the motherboard
- Non-volatile memory
- Used to store the computer’s BIOS (Basic Input/Output System), which contains the instructions for booting the computer

Slide 12: CACHE MEMORY

- High-speed memory between the CPU and main memory.
- Temporarily stores frequently used data and instructions for faster access.
- Small capacity, but faster and more expensive than main memory.
- **L1:** Fastest and smallest; usually inside the CPU (about 64 KB).
- **L2:** Larger than L1; on the CPU or a separate chip (about 1-2 MB).
- **L3:** Larger and slower than L1/L2, but faster than main memory.

Slide 13: SECONDARY STORAGE DEVICES

- Store data and programs long-term, even when the computer is turned off.
- **HDDs:** High-capacity magnetic storage using spinning platters.
- **SSDs:** Fast flash storage with no moving parts.
- **NVMe SSDs:** High-speed SSDs that connect over PCIe; common in modern computers.
- **Portable SSDs:** Compact external storage, often connected via USB-C.
- **Cloud storage:** Files stored on remote servers and accessed over the internet.

Slide 14: STORAGE CAPACITIES

- The main unit of storage is the binary digit or bit.
- 1 B = 1 $Byte$ = 8 $bits$
- 1 KB = 1 $KiloByte$ = 210 B = 1024 B
- 1 MB = 1 $MegaByte$ = 220 B = 1024 KB
- 1 GB = 1 $GigaByte$ = 230 B = 1024 MB
- 1 TB = 1 $TeraByte$ = 240 B = 1024 GB

Slide 15: FILE SYSTEMS, FOLDERS & FILE EXTENSIONS

- A **file system** organizes, names, and stores files on a storage device (for example, NTFS, APFS, or ext4).
- Folders (directories) group files and other folders in a hierarchy, starting from a root folder.
- A **path** shows where a file is stored, such as `Documents/Classes/notes.pdf`.
- A file extension usually indicates its format or associated application:
  - **Documents:** `.txt`, `.pdf`, `.docx`
  - **Images:** `.jpg`, `.png`, `.gif`
  - **Audio/video:** `.mp3`, `.mp4`
  - **Data/code:** `.csv`, `.html`, `.js`, `.py`
- Renaming an extension does not convert the file's actual format.

Slide 16: GRAPHICAL PROCESSING UNIT

- A GPU (Graphics Processing Unit) performs many operations in parallel.
- It renders graphics and accelerates video editing, scientific computing, and AI workloads.
- **Integrated GPU:** Built into a processor or system-on-chip; shares system memory and uses less power.
- **Discrete GPU:** Separate chip/card with dedicated video memory (VRAM); offers more performance.
- Modern GPUs may include specialized hardware for AI and ray tracing, plus features such as AI upscaling.
- Examples: NVIDIA GeForce RTX, AMD Radeon RX, and Intel Arc.

Slide 17: COMPUTER NETWORKS

- A computer network connects devices so they can communicate and share data and resources.
- Early networks, including ARPANET (1969), connected computers at research institutions.
- Different networks needed common rules to communicate; the TCP/IP protocol suite made this possible and underpins the internet.
- Today, networks use technologies such as Ethernet, Wi-Fi, and 4G/5G to connect computers, phones, and IoT devices.
- Networks enable services such as web browsing, messaging, video calls, cloud services, and online collaboration.

Slide 18: COMPUTER NETWORK ARCHITECTURE

- **End systems (hosts):** Devices that run network applications and send or receive data, such as computers, phones, and servers.
- **Packets:** Data is divided into small units for transmission across the network.
- **Packet switches:** Forward packets toward their destination. Routers connect networks; switches commonly connect devices within a LAN.
- **Communication links:** Carry packets between devices using wired media (such as Ethernet or fiber) or wireless signals (such as Wi-Fi or cellular).
- **Protocols:** Shared rules that define how devices format, send, route, and receive data.

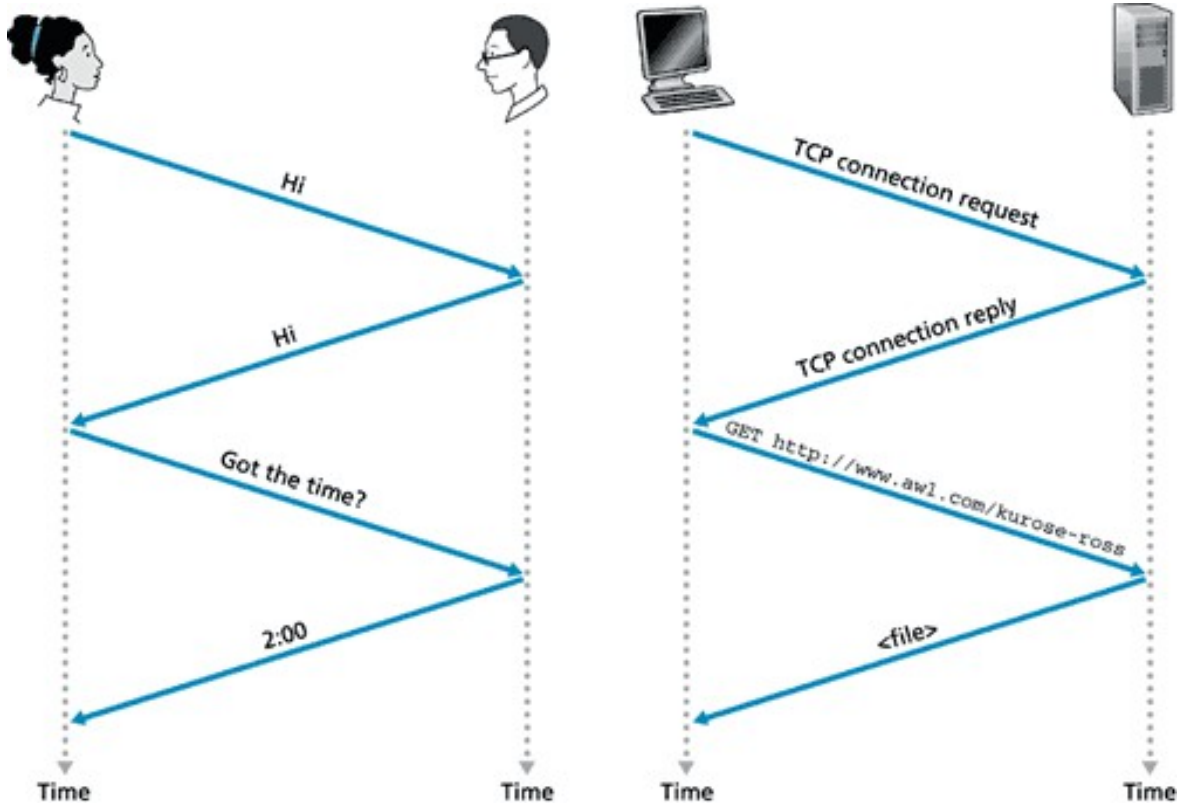
Slide 19: NETWORK HARDWARE & TYPES

Common network hardware

- **Network interface card (NIC):** Connects a device to a network.
- **Switch:** Connects devices within a local network.
- **Router:** Connects different networks and directs data between them.
- **Access point:** Provides wireless network access.
- **Modem and transmission media:** Connect to an internet service; carry data over cable, fiber, or radio.

  Network types by coverage
  - **LAN:** Local area, such as a home, school, or office.
  - **MAN:** Covers a city or metropolitan area.
  - **WAN:** Spans large regions or countries; the internet is the largest example.

Slide 20: Protocols



TCP/IP Protocol

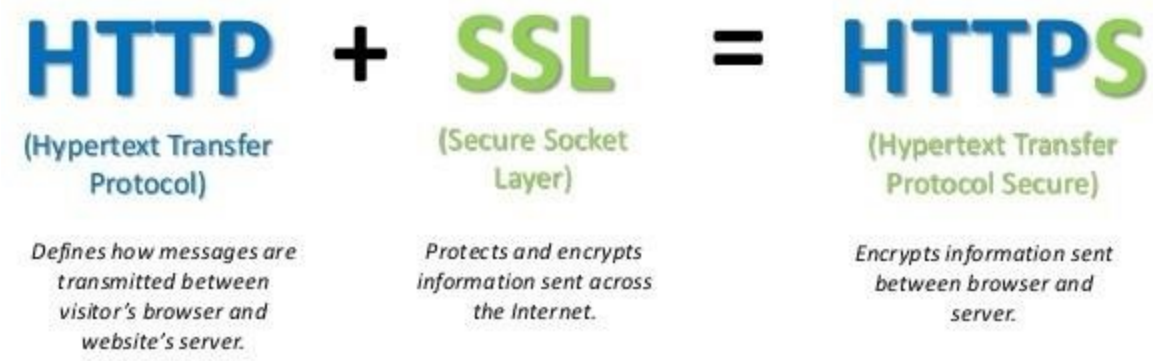
Typical protocols:

- DNS - name lookup
- DHCP - network configuration
- SMTP/IMAP - email
- FTP/SFTP - file transfer
- SSH - secure remote access

HTTP/S Protocol

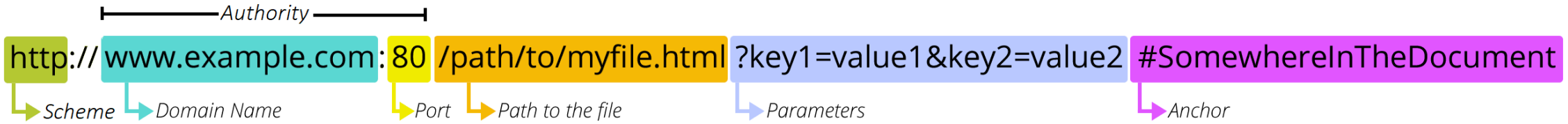
Slide 21: HTTP & HTTPS

- **HTTP (Hypertext Transfer Protocol):** Transfers web page data between a browser and a web server without encryption.
- **HTTPS (HTTP Secure):** Uses TLS to encrypt the connection, help prevent data tampering, and verify the server's identity.
- HTTPS helps protect information in transit, such as passwords and payment details.
- A padlock or `https://` does not guarantee that a website itself is trustworthy.



Slide 22: THE INTERNET

- The Internet is a global network of interconnected networks that communicate using the TCP/IP protocol suite.
- It is decentralized: no single organization owns or controls the entire Internet.
- Internet service providers, businesses, governments, and other organizations operate parts of its infrastructure.
- It enables services such as the web, email, messaging, video calls, and cloud applications.



Slide 23: DOMAIN NAME

- A **domain name** is a human-readable name used to find an internet service, instead of typing its IP address.
- **DNS (Domain Name System)** looks up domain names and returns associated records, often including IP addresses.
- In `https://www.example.com/products`, the hostname is `www.example.com`; `example.com` is the registered domain.
- `www` is a subdomain, `example` is the domain label, and `.com` is the top-level domain.
- A URL can also include a scheme (`https://`), a path (`/products`), and other parts.

Slide 24: INTERNET PROTOCOL ADDRESS

- An **IP address** is a numerical address assigned to a device or network interface so data can be routed to it.
- **IPv4** addresses look like `192.168.1.10`; **IPv6** addresses are longer and support many more devices.
- Addresses may be **public** (routable on the internet) or **private** (used within a local network).
- An IP address identifies a network connection, not a person's exact physical location, and it may change over time.

Slide 25: WEB BROWSER

Web Browser

- A web browser is an application for accessing websites and the Internet. When a user requests a web page from a particular website, the browser retrieves its files from a web server and then displays the page on the screen.
- Internet Explorer (IE)
- Mozilla Firefox
- Google Chrome
- Opera
- Safari
- Microsoft Edge

Slide 26: BROWSER PERMISSIONS & SECURITY

- **Review site permissions:** Allow or block access to your camera, microphone, location, and notifications for each website.
- **Grant access only when needed:** Remove permissions from sites that no longer need them.
- **Check security warnings:** Do not ignore browser warnings about unsafe sites or invalid certificates; HTTPS alone does not prove a site is trustworthy.
- **Protect privacy:** Block unwanted pop-ups, manage cookies and tracking settings, and clear browsing data when appropriate.
- **Stay safer:** Keep the browser up to date, use its built-in Safe Browsing protection, and install only trusted extensions.

Slide 27: BROWSER DEVELOPER TOOLS (DEVTOOLS)

- Built-in tools for inspecting, debugging, and testing web pages.
- **Elements/Inspector:** Examine page structure and styles; test temporary changes.
- **Console:** View errors and run JavaScript commands.
- **Network:** Inspect requests, responses, and loading times.
- **Sources/Debugger:** Set breakpoints and step through JavaScript.
- **Performance:** Diagnose page speed and runtime issues.
- **Application:** Inspect site storage such as cookies, local/session storage, and caches; manage service workers and site data.
- Often opened with **F12** or **Ctrl+Shift+I** (Windows/Linux); shortcuts vary by browser.

Slide 28: VIRTUAL PRIVATE NETWORK

- A **VPN (Virtual Private Network)** creates an encrypted connection, or tunnel, between your device and a VPN server.
- It can protect traffic on untrusted networks and provide secure access to an organization's private network.
- Websites generally see the VPN server's IP address instead of your device's public IP address.
- A VPN does **not** make you anonymous: the VPN provider may see connection information, and websites can still track you.
- Use HTTPS as well; a VPN does not replace secure website connections or protect against malware and phishing.

Slide 29: FIREWALL

- A **firewall** monitors and filters network traffic according to security rules.
- **Host-based firewalls** protect an individual computer; **network firewalls** control traffic between networks.
- Rules can allow or block connections based on factors such as addresses, ports, protocols, and connection state.
- Firewalls help reduce unauthorized access, but do not replace software updates, secure passwords, or protection against phishing.